

Quantifying AI Over-Reliance in Programming Education: A Telemetry-Driven Behavioral Metric with Productive Friction in S-SPARC

S-SPARC & E-STRANGE Research Initiative

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Abstract: While generative artificial intelligence enables students to produce functional code instantly, unconstrained access fosters cognitive offloading, surface-level self-regulation, and epistemic dependence. This paper presents an empirical framework to quantify, diagnose, and mitigate AI over-reliance in computing education. We formulate the **AI Over-Reliance Index (ORI)**, a composite behavioral metric synthesizing four unobtrusive telemetry streams: Shannon information entropy (S_{prompt}), structured problem-framing completeness (S_{CIOE}), conceptual inquiry ratio (S_{concept}), and keystroke-verified productive friction adherence (S_{reflect}). Operating natively within the S-SPARC engine and E-STRANGE auto-grader, the system records continuous millisecond-level telemetry with zero survey administration overhead. To establish construct validity while circumventing circular estimation, we detail a two-phase psychometric roadmap: Phase 1 ($N = 35 - 50$) conducts Exploratory Factor Analysis (EFA) to discover latent factor structures, while Phase 2 ($N \geq 80$) conducts independent Confirmatory Factor Analysis (CFA) and convergent validation against the standardized GenAI Reliance Scale. Key threats to validity, including maturation confound, entropy inversion, classifier error propagation, and Pygmalion bias, are formally analyzed and mitigated.

Index Terms: Artificial intelligence in education, cognitive offloading, educational data mining, information entropy, learning analytics, productive friction, programming pedagogy.

I. INTRODUCTION

The widespread availability of Large Language Models (LLMs) has fundamentally altered software engineering pedagogy. Students increasingly leverage automated conversational assistants to write, debug, and optimize programming exercises. However, emergent computing education research reveals a growing pedagogical paradox: while automated code generation elevates short-term task completion rates, unassisted problem-solving ability deteriorates substantially [2], [3].

In computing laboratories, unconstrained AI access frequently produces a passive behavioral cycle: students copy problem specifications directly into the model, extract generated code without cognitive verification, and submit the output to automated graders. Because test suites evaluate only functional correctness, students receive unearned credit while bypassing the critical cognitive processes of mental tracing, syntax structuring, and edge-case synthesis. This phenomenon, termed *cognitive atrophy*, severely impairs independent programming capability during unassisted examinations [2], [6].

To resolve this challenge, educators require quantitative, real-time diagnostic instruments that can differentiate productive AI-assisted inquiry from epistemic over-reliance. This paper details the mathematical formulation, system architecture, and empirical validation design of the AI Over-Reliance Index (ORI) implemented within the S-SPARC pedagogical framework.

II. RELATED WORK & RESEARCH GAP

Prior literature evaluating student-AI interaction primarily relies on three distinct paradigms:

A. Post-Hoc Self-Report Surveys: Standardized instruments, such as Likert-scale questionnaires [2], [7], evaluate student perceptions of AI reliance. While psychometrically valuable, self-reports suffer from recall bias, social desirability, and a

fundamental disconnect from active coding behavior during laboratory tasks.

B. Binary Plagiarism Detection: Machine learning text classifiers attempt to identify AI-generated submissions. However, these tools exhibit high false-positive rates and operate with a purely punitive objective, providing zero diagnostic feedback regarding student reasoning processes.

C. Controlled Laboratory Studies: Recent investigations have utilized eye-tracking and qualitative dialogue coding to characterize reliance taxonomies [15], [16]. Although empirically rigorous, these methods cannot be computed continuously in live classroom deployments.

As summarized in Table I, S-SPARC bridges this gap by computing an unobtrusive, continuous composite metric directly coupled with LMS unit-test execution.

TABLE I: COMPARATIVE PARADIGM ANALYSIS

Methodology	Primary Limitation	S-SPARC Advantage
Likert Surveys [2], [7]	Recall bias; subjective perception	Live millisecond telemetry
AI Text Detectors	High error; punitive only	Cognitive inquiry profiling
Eye-Tracking [16]	Post-hoc; lab only	Real-time live metric (ORI)
Commercial AI [5]	Unchecked prompt spam	Productive friction (60s)

Comparison of reliance assessment paradigms in higher CS education.

III. MATHEMATICAL METRIC FORMULATION

The AI Over-Reliance Index (ORI) is formulated as an inverse composite index bound on the domain $[0, 1]$, where values approaching 1.0 denote severe cognitive dependence, and values approaching 0.0 denote autonomous, reflective inquiry:

$$\text{ORI} = 1 - \sum_{i=1}^4 w_i \cdot X_i \quad (1)$$

subject to $\sum w_i = 1.0$ and $X_i \in [0, 1]$. The four indicator components are operationalized as follows:

1) *Shannon Information Entropy* ($S_{Q_{\text{prompt}}}$): Adapting information theory [11], we evaluate the lexical diversity and technical syntax density of student prompts:

$$H(X) = - \sum_{j=1}^n p(x_j) \log_2 p(x_j) \quad (2)$$

where $p(x_j)$ represents the relative frequency of programming keywords, variable identifiers, and logic tokens. High entropy reflects granular, context-rich problem specification, penalizing generic queries (e.g., "why code error") [13], [17]. Initial baseline weight: $w_1 = 0.30$.

2) *Context Completeness Score* (S_{CIOE}): Evaluates prompt structural rigor across four dimensions: Context ($\mathcal{C}$), Input constraints ($\mathcal{I}$), Expected Output ($\mathcal{O}$), and Error Trace ($\mathcal{E}$) [12]. Normalization follows:

$$S_{\text{CIOE}} = \frac{1}{4} (\mathcal{C} + \mathcal{I} + \mathcal{O} + \mathcal{E}) \quad (3)$$

where each dimension is evaluated as a boolean indicator. Initial baseline weight: $w_2 = 0.25$.

3) *Conceptual Inquiry Ratio* ($S_{R_{\text{concept}}}$): Quantifies the proportion of queries targeting algorithmic logic and mental models (Bloom Levels C2-C4) versus direct code generation requests (Bloom C1):

$$S_{R_{\text{concept}}} = \frac{N_{\text{conceptual}}}{N_{\text{conceptual}} + N_{\text{direct_code}}} \quad (4)$$

Initial baseline weight: $w_3 = 0.20$.

4) *Productive Friction Adherence* ($S_{F_{\text{reflect}}}$): Enforces a mandatory 60-second reflection interval [9] between AI responses and code compilation attempts. To prevent meta-gaming via external LLMs, $S_{F_{\text{reflect}}}$ integrates keystroke cadence verification (Δt_{dwell}):

$$S_{F_{\text{reflect}}} = \min\left(1.0, \frac{t_{\text{pause}}}{60}\right) \cdot \text{cadence}_{\text{manual}} \quad (5)$$

Initial baseline weight: $w_4 = 0.25$.

IV. DATA ARCHITECTURE & OPERATIONAL PIPELINE

The telemetric logging subsystem is natively integrated into the MySQL backend of S-SPARC and E-STRANGE. Table II summarizes the core telemetry schema captured passively per student interaction.

TABLE II: RELATIONAL TELEMETRY SCHEMA

Variable	Measurement Granularity
prompt_raw_text	UTF-8 text, timestamp (ms)
entropy_score	Float $[0.0, 5.0]$, token frequency
cioe_vector	Bitmask $\angle \mathcal{C}, \mathcal{I}, \mathcal{O}, \mathcal{E} \in \{0, 1\}^4$
keystroke_dwell	Inter-keystroke variance (ms)
test_pass_rate	Passed test ratio $[0.0, 1.0]$
carbon_grams	Model token compute estimation

V. METHODOLOGICAL DEFENSES & THREATS TO VALIDITY

A. Reactivity and Maturation Confound: A single-cohort A-B intervention (Baseline Modules 1-2 without friction vs. Intervention Modules 3-4 with friction) introduces a potential maturation confound, wherein temporal learning coincides with system intervention. In our manuscript, this limitation is formally acknowledged. When two parallel laboratory sections are available, a counterbalanced crossover design (Section 1: A-B; Section 2: B-A) is executed to isolate temporal maturation from intervention efficacy.

B. Two-Phase Psychometric Roadmap: Validating latent dimensionality across 4 indicators requires distinct samples to avoid structural overfitting:

- **Phase 1 (Cohort 1, $N = 35 - 50$):** Exploratory Factor Analysis (EFA) to discover factor loading structures and calibrate empirical regression weights.
- **Phase 2 (Cohort 2, $N \geq 80$):** Confirmatory Factor Analysis (CFA) to confirm unidimensionality and convergent validity against the standardized GenAI Reliance Scale (Hou et al. 2025 [1]).

C. Entropy Inversion Paradox: Expert programmers frequently author concise, highly specific prompts with low lexical redundancy, whereas struggling novices may generate rambling, unfocused text. To prevent expert conciseness from being misclassified as novice over-reliance, we model an explicit interaction term ($\text{Entropy} \times \text{Baseline Ability}$) using pre-test scores and GPA as covariates.

D. NLP Classifier Validation: Automated classification of $S_{R_{\text{concept}}}$ intent is subjected to a formal validation sub-study. Double annotation by two independent domain experts across ≥ 100 prompt samples must achieve an inter-rater agreement of Cohen's Kappa $\kappa \geq 0.80$ prior to metric integration.

E. Pygmalion Effect & Ethical Governance: To prevent algorithmic labeling from biasing instructor grading, individual ORI scores are strictly blind-masked from instructors during Phase 1. Full dashboard analytics unlock in Phase 2 only after psychometric validation is complete.

VI. STATISTICAL INTEGRITY & OPEN SCIENCE

To eliminate researcher degrees of freedom and circularity, the research protocol enforces:

1) *5-Fold Cross-Validation:* Regression weights are calibrated on training folds and evaluated on held-out validation folds.

2) *Multicollinearity Testing:* Predictor variance inflation factors are verified ($\text{VIF} < 5.0$) to prevent parameter instability.

3) *OSF Pre-Registration:* Analysis scripts and confirmatory hypotheses are locked on the Open Science Framework (OSF.io) prior to data collection.

4) *Family-Wise Error Control:* Four secondary exploratory hypotheses (Green AI volume paradox, AST distance delta, reflexive pause pass rates, and semantic cache maturity) are evaluated under Bonferroni-adjusted significance ($\alpha_{\text{adjusted}} = 0.05 / 4 = 0.0125$).

VII. IMPLEMENTATION TIMELINE & ROADMAP

The implementation timeline spans two academic semesters:

TABLE III: IMPLEMENTATION TIMELINE

Phase / Period	Key Research Activities	Deliverable
Phase 1 (W1-2)	OSF Pre-Registration, Baseline logging	Locked protocol
Phase 1 (W3-4)	Intervention Phase (60s friction)	Telemetry dataset
Phase 1 (W5-6)	NLP annotation ($\kappa \geq 0.80$), EFA	Factor report
Phase 1 (W7-8)	Conference drafting (IEEE TALE)	Submitted paper
Phase 2 (Sem 2)	CFA replication ($N \geq 80$), Scopus Q1	IEEE TLT Paper

VIII. CONCLUSION & SCOPE BOUNDARY

The S-SPARC AI Over-Reliance framework provides an empirical, telemetry-driven methodology to diagnose and mitigate cognitive offloading in programming education. The scope of this metric is formally bounded to deterministic unit-testing environments (Computer Science); generalization to non-computing disciplines represents future work.

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